

SRM VEC
DEPARTMENT OF AIDS
Semester End Examination
(Regulations 2023)
AD3445 – AI1

Branch: B.Tech | Date: 01.02.2026 | Max. Marks: 100 | Duration: 3 Hrs

Answer ALL Questions
(100 Marks)

PART A — (10 × 2 = 20 marks)

1. Define a Rational Agent and specify the four factors that determine rationality. [CO1] [BTL1-Remember] (2)
2. Differentiate between Informed and Uninformed search strategies with one example each. [CO1] [BTL2-Understand] (2)
3. What is the role of a heuristic function in A* search? State the condition for optimality. [CO1] [BTL1-Remember] (2)
4. Explain the concept of 'Constraint Propagation' in CSP. [CO2] [BTL2-Understand] (2)
5. Define Alpha-Beta pruning. Why is it used in game playing? [CO2] [BTL1-Remember] (2)
6. Convert the following sentence into First Order Predicate Logic (FOPL): 'Every student who takes AI likes logic.' [CO3] [BTL3-Apply] (2)
7. Distinguish between Forward Chaining and Backward Chaining. [CO3] [BTL2-Understand] (2)
8. State the Bayes' Rule and explain each term. [CO4] [BTL1-Remember] (2)
9. What is a Markov Blanket in a Bayesian Network? [CO4] [BTL2-Understand] (2)
10. Define 'Information Gain' in the context of Decision Tree learning. [CO5] [BTL1-Remember] (2)

PART B — (5 × 16 = 80 marks)

11(a) (i) Explain the A* search algorithm. Prove that A* is optimal if the heuristic function is admissible. (ii) Describe the PEAS specification for an Automated Medical Diagnosis System.

OR

(b) (i) Compare Breadth-First Search and Depth-First Search based on Time Complexity, Space Complexity, Completeness, and Optimality. (ii) Illustrate the working of Hill Climbing search and explain the 'Local Maxima' problem with a diagram.

12(a) Consider a game tree where the root is a MAX node. Apply the Minimax algorithm with Alpha-Beta pruning to find the optimal move. Assume the leaf nodes (left to right) are: 3, 12, 8, 2, 4, 6, 14, 5, 2. Show the pruned branches and the final value at the root.

OR

(b) (i) Define Constraint Satisfaction Problem (CSP). Explain the Backtracking Search algorithm for CSP. (ii) Solve the following Cryptarithmic problem: SEND + MORE = MONEY. Show the steps and constraints.

13(a) (i) Explain the process of Unification in First Order Logic with a suitable example. (ii) Use Resolution to prove that 'Fido will die' given: 1. All dogs are mammals. 2. All mammals are animals. 3. Fido is a dog. 4. All animals will die.

OR

(b) (i) Describe the various steps involved in converting a FOPL sentence into Conjunctive Normal Form (CNF). (ii) Explain the Frame Problem in Knowledge Representation.

14(a) Construct a Bayesian Network for the 'Burglary' problem (Burglary, Earthquake, Alarm, JohnCalls, MaryCalls). (i) Draw the network topology. (ii) Explain the conditional probability tables (CPTs). (iii) Calculate $P(B | A)$ given the network parameters.

OR

(b) (i) Explain the Hidden Markov Model (HMM) and its three fundamental problems (Evaluation, Decoding, Learning). (ii) Discuss the Axioms of Probability and their importance in AI.

15(a) Explain the ID3 algorithm for Decision Tree induction. Using a sample dataset of 'Weather' (Outlook, Temp, Humidity, Wind) and target 'PlayTennis', demonstrate how the root node is selected using Entropy and Information Gain.

OR

(b) (i) Describe the architecture of a simple Artificial Neural Network (Perceptron) and explain the backpropagation learning rule. (ii) Briefly explain Reinforcement Learning with the help of the Agent-Environment interaction loop.